

Pranav P

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Profile

Cybersecurity and AI enthusiast with hands-on experience in network security, Linux systems, and intelligent data-driven solutions. Passionate about securing digital infrastructure, analyzing threats, and developing practical solutions to real-world cybersecurity challenges. Strong collaborator with leadership experience, problem-solving ability, and a keen interest in offensive and defensive security practices.

Skills

Programming Languages: Python, Java, C, SQL

Cybersecurity: Network Security, Vulnerability Assessment, Penetration Testing Fundamentals, Digital Forensics, Linux Systems.

Tools & Technologies: Wireshark, Nmap, Linux.

Soft Skills: Leadership, Team Management, Analytical Thinking, Problem Solving, Communication

Education

APJ Abdul Kalam Technological University

September 2025 – June 2027

M.Tech in Cyber Forensics and Information Security

ER&DCI Institute of Technology

- Current CGPA: 7.81/10

- **Coursework:** Information Security, Incident Response, Cyber Forensics, Ethical Hacking, Network Security

APJ Abdul Kalam Technological University

Nov 2021 – June 2025

B.Tech in Artificial Intelligence and Data Science

St Joseph's College of Engineering and Technology, Palai

- Final CGPA: 7.84/10

- **Coursework:** Machine Learning, Deep Learning, NLP, Data Science, Big Data Analytics, DSA, OOPs

Sree Buddha Central School, Karuanagappally (CBSE)

2019 – 2021

Higher Secondary Education

- Grade: 90.8%

High School Education

2019

- Grade: 95.6%

Publications

Automated Mark Entry in Educational Institutions Using Multiple CNNs: A Case Study on Recognizing Handwritten Fractional Marks

Published in 2024 International Conference on Computational Intelligence and Network Systems (CINS), Dubai, UAE.

DOI: 10.1109/CINS63881.2024.10864430

Projects

Marks2CSV

github.com/MakersCircle/marks2csv

- Developed an AI-based system to automate the recognition and conversion of handwritten marks on educational answer sheets into structured CSV files. Utilized a combination of three CNN models to accurately identify single-digit and fractional marks, reducing manual data entry time and errors significantly. Led the project and designed the system and models
- Tools and Concepts Used: TensorFlow, openCV, Streamlit, Latex, Convolutional Neural Networks (CNN)

Drowsiness Detection System

github.com/pranavxp/Drowsiness

- Machine learning project for self skill boost, this is a project aimed towards drivers, by analyzing key facial features, particularly eye movements and blink patterns and immediately alert the drivers.
- Technologies and Concepts Used: OpenCV, Dlib, Eye Aspect Ratio (EAR)

Accident Anticipation System

github.com/MakersCircle

- Designed and implemented a real-time accident anticipation program using computer vision and deep learning techniques to predict traffic accidents from dash cam footage.

- Technologies and Concepts Used: Python, TensorFlow/PyTorch, Monocular Depth Estimation, Graph Neural Networks (GNNs), Temporal Attention, Multi-Task Learning

Intelligent VPN Leak Detection Using Network Traffic Classification

Cybersecurity Project

- Developed an intelligent VPN leak detection system to identify possible traffic leaks by analyzing network flow behavior instead of relying only on basic VPN ON/OFF checks.
- Implemented a machine learning based traffic classification approach to distinguish VPN-like and non-VPN-like flows, helping detect partial, intermittent, or application-specific VPN leakage.
- Designed the system to process captured network traffic, extract flow-level features, score VPN probability, and flag suspicious leak-like connections for further analysis.
- Tools and Concepts Used: Python, Scikit-learn, Random Forest, Network Traffic Analysis, Flow Feature Extraction, VPN Leak Detection, Cybersecurity

Courses and Certificates

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| • The Joy Of Computing Using Python | <i>IIT Madras (NPTEL)</i> |
| • Fundamentals Of Artificial Intelligence | <i>IIT Guwahati (NPTEL)</i> |
| • Wheeled Mobile Robots | <i>IIT Madras (NPTEL)</i> |
| • Google Cybersecurity | <i>Google (Coursera)</i> |